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An Identifiability-First Feasibility Protocol for a Gaussian-Regularized Diósi Collapse Test with a Casimir-Boundary Control

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> **Scientific standing.** This article does not report measured evidence for objective collapse, a physically realized mesoscopic superposition, a Casimir-to-collapse coupling, or spacetime-manifold dynamics. It reports a model-specific, synthetic identifiability result and defines the measurements required to determine whether that result survives in hardware.

Abstract

We present an identifiability-first protocol for testing one specified collapse dynamics: the nondissipative, Gaussian-regularized Diósi mass-density model applied to a single effective particle. Penrose objective reduction supplies motivation for asking whether spatially distinct mass-density branches have a finite lifetime; it does not supply the stochastic master equation used for the forecast. A nearby conducting boundary is treated as a secondary, independently controlled quantum-electrodynamic perturbation. No Casimir variable enters the registered collapse generator.

The methodological result is a design progression ending in a bounded leading design. An initial synthetic apparatus was rejected because its candidate collapse signature was nearly collinear with modeled nuisance responses. A first bounded redesign established a usable covariance-whitened geometry, with maximum signature cosine 0.7177 and normalized Gram condition number 6.53, but its 0.9979 power and 542-window forecast belonged to that parent synthetic world. A subsequent 200-candidate apparatus search retained the same whitened geometry and frozen collapse law while applying mass-density, phase, gas, preparation-scale, external-bound, power, and companion gates. Three candidates survived. The leading candidate requires 1,028 paired windows and has forecast power 0.927 at 1,600 windows. None of these power values is empirical.

A downstream synthetic diagnostic now makes the boundary question explicit: a normalized four-cell complex cross-ratio tests boundary--branch nonfactorization while canceling the boundary-independent registered Diósi factor. Its null recovery and injected-signal tests pass, but no measured branch-control, wave-packet, ordinary-response, or interaction data exist.

The leading design is now also bound to a material/Green/FDT ordinary-null runtime. In its synthetic recovery fixture, the finite-geometry response gives an electromagnetic phase of 0.0200 rad and an echo contraction exponent of 1.34×10^{-6}; zero-coupling, infinite-distance, detailed-balance, Planck--Stefan--Boltzmann, and intervention-recovery gates pass. These are software-recovery values, not apparatus forecasts: the specimen spectrum, as-built geometry, full-Maxwell solution, independent solver check, and measured response covariance remain absent.

A constituent and environmental diagnostic separates Hamiltonian phase from nonunitary contraction and stress-tests the mass model. The leading synthetic commissioning design uses a diamond-density 276.302 nm-radius sphere with mass 3.09251×10^-16 kg, 250 nm tangential branch separation, 250 ms hold time, 10 µm boundary gap, 4 K temperature, and 10^-15 Pa pressure. At the selected regularization length R0=100 nm, the registered effective-particle model predicts a conditional rate of 0.118046 s^-1 and 2.908% DP-only visibility

loss. Transporting the registered mass-density sensitivity envelope lowers the design floor to 0.435%. The ordinary-background surrogates yield gas-to-DP ratio 0.00732 and echoed phase uncertainty 1.11×10^-8 rad, but they remain unmeasured. The design is therefore a commissioning target, not a physically admitted apparatus.

The proposal remains an empirical no-go until the stated mass can be prepared in a verified spatial superposition, finite-geometry electromagnetic and material responses are measured, the covariance model passes preregistered stress tests, and an interpretation-eligible companion observable is demonstrated. The contribution is therefore a falsifiable protocol and a transparent map of its dominant failure modes, not evidence that objective collapse occurs.

Keywords: objective collapse; Diósi model; Penrose objective reduction; matter-wave coherence; Casimir boundary; identifiability; covariance whitening; experimental design

1. Scientific question and result

The experiment asks a narrow question:

> After ordinary thermal, electromagnetic, gas, vibration, readout, and boundary-correlated responses have been measured, can the remaining complex coherence data support or exclude the frozen mass–separation–hold-time signature of one regularized Diósi model?

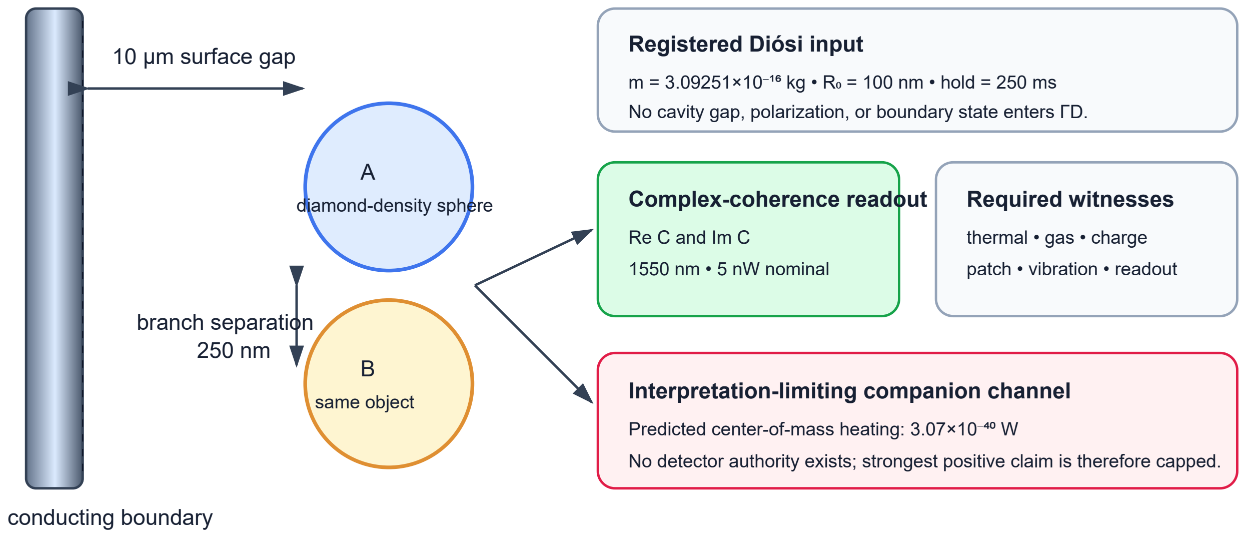
This question is intentionally narrower than “does gravity collapse the wave function?” A null result can constrain only the registered generator and parameter point within the experiment's demonstrated sensitivity. A positive residual can be assigned to that generator only if it is identifiable against the nuisance span and if the model's independently registered companion prediction is observed. Neither outcome, by itself, establishes Penrose's spacetime interpretation.

The present result is conditional:

1. a first synthetic design is non-identifiable and is rejected;
2. a bounded parent redesign establishes a usable nominal whitened geometry;
3. three of 200 downstream configurations pass every transported synthetic gate;
4. the leading configuration predicts 2.908% effective-Gaussian loss and a 0.435% conservative density-envelope loss at 250 ms;
5. its selected-candidate acquisition forecast is 1,028 paired windows and power 0.927 at 1,600 windows;
6. no measured response, covariance, state-preparation, or companion-detector receipt yet authorizes a physical pilot or confirmatory campaign.

Figure 1 gives the proposed geometry. Figure 2 shows the acquisition sequence.

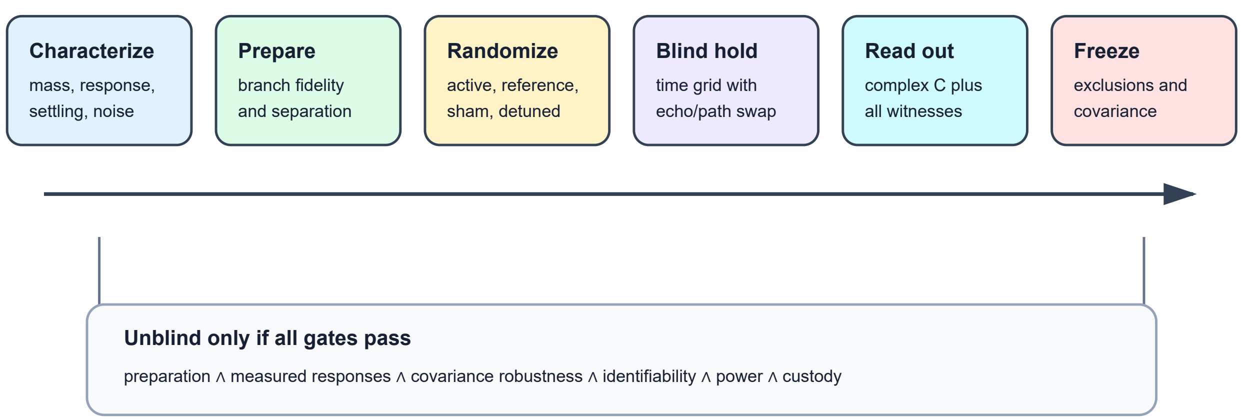
Leading synthetic commissioning design



Branches are drawn parallel to the boundary so the nominal gap is common. Diagram not to scale.

Figure 1. The boundary is an electromagnetic control, not a term in the registered Diósi generator. The two center-of-mass branches are separated parallel to the boundary so that the nominal surface distance is common to both branches. Environmental witnesses and a separate companion channel are required for interpretation.

Pilot and confirmatory timing logic



A fixed “0.5 Hz” label does not determine settling. Static randomized cells are the default until the measured transfer function authorizes a cadence.

Figure 2. Preparation and branch verification precede the blinded hold. Path swap, echo, sham-switch, and detuned-boundary cells are randomized. Unblinding is permitted only after exclusions, response vectors, covariance ancestry, and analysis code are frozen.

2. Exact tested dynamics

2.1 Registered Gaussian-regularized Diósi model

The executable prediction is not a generic "Diósi–Penrose" rate. It is one named reduced-order implementation: a nondissipative Diósi mass-density master equation for a single effective particle whose mass-density operator is Gaussian-smeared over the physical length R0 [1–3]. In schematic form,

$$\frac{d\hat{\rho}}{dt} = -\frac{i}{\hbar}[\hat{H},\hat{\rho}] - \frac{G}{2\hbar}\int d^3x\, d^3y\, \frac{[\hat{M}_{R_0}(\mathbf{x}),[\hat{M}_{R_0}(\mathbf{y}),\hat{\rho}]]}{|\mathbf{x}-\mathbf{y}|}. \tag{1}$$

The implementation freezes the convention, the mass representation, and R0. For two branches of the same effective Gaussian particle, with mass m and center separation d, the registered self-energy difference is

$$E_G(m,d,R_0) = Gm^2\left[\frac{1}{\sqrt{\pi}R_0} - \frac{\mathrm{erf}(d/2R_0)}{d}\right], \qquad \Gamma_{\mathrm{D}} = \frac{E_G}{\hbar}, \qquad \mathcal{V}_{\mathrm{D}}(t) = e^{-\Gamma_{\mathrm{D}}t}. \tag{2}$$

The d→0 limit is evaluated analytically. A Fourier-space Simpson quadrature is used only as a numerical cross-check; its softening parameter is not the physical cutoff.

For the leading synthetic commissioning point,

$$\begin{aligned} m &= 3.0925053\times10^{-16}\ \mathrm{kg},\\ d &= 2.50\times10^{-7}\ \mathrm{m},\\ R_0 &= 1.00\times10^{-7}\ \mathrm{m},\\ E_G &= 1.2448784\times10^{-35}\ \mathrm{J},\\ \Gamma_{\mathrm{D}} &= 1.1804586\times10^{-1}\ \mathrm{s}^{-1},\\ \tau_{\mathrm{D}} &= 8.47128\ \mathrm{s}. \end{aligned} \tag{3}$$

At t=0.25 s,

$$1-\mathcal{V}_{\mathrm{D}} = 1-e^{-\Gamma_{\mathrm{D}}t} = 2.90803\times10^{-2} = 2.90803\%. \tag{4}$$

This is the registered effective-Gaussian forecast for the leading design. A second value is required because the mass-density representation is not yet an empirical authority. Transporting the Stage-4.2L representation envelope gives an exponent of 0.00435852, or a conservative 0.434903% loss at the same hold time. The paper therefore carries 2.90803% as the named-model prediction and 0.434903% as the design floor; neither is a total measured-visibility forecast.

2.1.1 From Schrödinger evolution to the measured residual

The experiment does not measure E_G directly. It reconstructs the off-diagonal center-of-mass coherence $C(t) = \langle A|\hat{\rho}(t)|B\rangle$. For two Hamiltonian energy eigenstates, ordinary Schrödinger evolution gives

$$|\psi(t)\rangle = c_1 e^{-iE_1t/\hbar}|E_1\rangle + c_2 e^{-iE_2t/\hbar}|E_2\rangle, \qquad \rho_{12}(t) = c_1c_2^* e^{-i(E_1-E_2)t/\hbar}.$$

Thus the beat frequency is $(E_1-E_2)/\hbar$, while $|\rho_{12}(t)| = |c_1c_2^*|$ remains constant in an ideal closed system. The energy variance $\sigma_H^2 = \langle H^2\rangle - \langle H\rangle^2$ describes the outcome spread of Hamiltonian-energy measurements; it is not the gravitational mass-density difference energy E_G. The localized apparatus branches

$|A\rangle$ and $|B\rangle$ need not themselves be stationary energy eigenstates, so the experiment reconstructs their projected complex coherence rather than assuming a literal two-line atomic spectrum.

For the registered hypothesis lanes, the transparent scalar form is

$$C(t) = C(0)e^{-i\Delta E_H t/\hbar}e^{-\chi_{\text{env}}(t)}e^{-E_G t/\hbar}. \tag{4a}$$

The Hamiltonian energy difference ΔE_H rotates phase. The ordinary open-system functional χ_{env} represents environmental contraction after the environment is traced out. The final factor is the additional nonunitary contraction of the frozen Diósi model. Equal units do not make these three objects interchangeable: in particular, a Schrödinger energy variance, a photon frequency, and the gravitational mass-density difference energy are not the same source term.

After ordinary loss has been estimated from blinded controls, a remaining scalar contraction can be expressed as

$$E_{\text{D,eq}} = -\frac{\hbar}{t}\ln\left(\frac{|C_{\text{residual}}(t)|}{|C(0)|}\right). \tag{4b}$$

Equation (4b) is a conditional inference under the registered exponential model, not a calorimetric measurement of gravitational energy. The full whitened complex estimator in Eqs. (8)--(9), rather than this scalar summary, remains authoritative for distinguishing phase, loss, and correlated nuisance responses.

2.1.2 Constituent mass density changes the forecast

Equation (2) treats the apparatus as one effective Gaussian particle. Stage 4.2J also evaluates a homogeneous sphere of physical radius R, convolved with the identical Gaussian regularization R_0. Its radial Fourier form is

$$E_G^{\text{sphere}} = \frac{2Gm^2}{\pi R_0}\int_0^\infty du\, e^{-u^2}\left[\frac{3(\sin q - q\cos q)}{q^3}\right]^2\left[1-\frac{\sin(ud/R_0)}{ud/R_0}\right], \qquad q=\frac{uR}{R_0}. \tag{4c}$$

For the earlier silica reference, the converged homogeneous-sphere diagnostic was 0.249896 of its effective-Gaussian energy. Stage 4.2L subsequently widened the registered representation envelope to a factor of 6.77099. Stage 4.2M transports that factor to the leading candidate rather than pretending that a diamond-density label supplies an atomistic density map. This yields the 0.434903% conservative design floor quoted above. A direct homogeneous, layered, coarse-grained, and atomistic calculation for the selected specimen remains blocked pending provenance-bound density and coating inputs.

2.1.3 Cheap feasibility screens precede power

For an ideal equilibrium residual gas, the Stage-4.2J conservative screen uses

$$n=\frac{P}{k_BT}, \qquad \bar{v}=\sqrt{\frac{8k_BT}{\pi m_g}}, \qquad \Gamma_{\text{coll}}^{\text{screen}} = n\bar{v}\,\pi R^2. \tag{4d}$$

The earlier 2×10^{-11} Pa silica reference failed this screen. The leading design therefore registers 4 K and 10^{-15} Pa. Its transported Stage-4.2L QLBE proxy gives $\Gamma_{\text{gas}} = 8.64118\times10^{-4}$ s^{-1}, or 0.007320 of the registered DP rate. This clears the synthetic one-tenth-DP gate, but it is not a measured vacuum receipt or a complete collisional decoherence kernel [13]. Species-resolved differential scattering, confinement, pressure calibration, and collision-veto performance must replace the proxy. The leading mass is also about 1.0955×10^6 times the 170 kDa cross-platform benchmark cited in Ref. [6], so state preparation remains the first independent hardware gate.

Hydrogen and Rydberg scales remain useful dimensional calibration checks. The registered E_G is about 5.71×10^{-18} of the Rydberg energy, but that ratio supplies neither a cavity coupling nor a collapse mechanism. It only states the scale of the residual that the coherence estimator would infer if the frozen Diósi model were selected after ordinary channels were excluded.

2.2 Penrose OR motivation, notation, and scope

Penrose's objective-reduction argument motivates an order-of-magnitude lifetime $\tau_{\mathrm{OR}} \sim \hbar/E_G$ for incompatible mass distributions [4,5]. That heuristic and Eq. (2) share a gravitational self-energy scale, but they are not the same theory object:

Question	Penrose OR motivation	Registered Diósi runtime
Why consider finite branch lifetime?	incompatibility of alternative spacetime geometries	not supplied
Numerical lifetime scale	$\tau \sim \hbar/E_G$	$\Gamma = E_G/\hbar$
Stochastic master equation	not uniquely specified	specified
Gaussian cutoff R0	not fixed by OR	frozen at 100 nm
Diffusion/heating companion	not uniquely specified	specified
Casimir-boundary modifier	not supplied	absent

Consequently, an exclusion of Eq. (2) is not an exclusion of every Penrose-motivated theory. A match to Eq. (2) is not direct evidence for spacetime “snapping” or manifold instability.

2.3 Mass-density representation is part of the hypothesis

The leading object's geometric radius is 276.3 nm, but Eq. (2) treats its center of mass as one Gaussian-smeared effective particle. This is not equivalent to resolving a selected specimen's atomic or layered density. The current representation audit is:

Representation of the same apparatus	Current standing	Consequence
single effective Gaussian particle	executable; Eq. (2)	2.90803% leading-design forecast
transported Stage-4.2L density envelope	executable sensitivity floor	0.434903% leading-design floor; not a selected specimen map
homogeneous selected sphere	not yet recomputed for the Stage-4.2M material identity	bulk-profile dependence unresolved
coated or layered selected sphere	not computed	coating dependence unknown
coarse-grained or atomistic selected density	no admissible density provenance	granularity dependence unknown
R0 sweep with converged selected representations	sensitivity-only calculations exist	parameter stability unresolved

Therefore this article proposes a test of the effective-particle Diósi model, not a representation-independent or generic DP test. A revised confirmatory analysis must either demonstrate stability across physically defensible mass representations or preserve that narrower claim in its title and conclusion.

3. Leading Synthetic Commissioning Design

The following manifest is the current target for measured subsystem commissioning. It supersedes earlier design assumptions for forward planning, but it does not claim an as-built apparatus or physical feasibility.

Field	Leading design value	Evidence role
identity	`stage4_2m_candidate_002`	frozen synthetic search identity
material and geometry	diamond-density sphere	bounded-search assumption; specimen not selected
radius	276.302 nm	derived from frozen mass/material identity
mass	3.0925053×10^-16 kg = 1.86235×10^11 Da	frozen model input
branch separation	250 nm, parallel to boundary	frozen model input
principal hold time	250 ms	frozen model input
hold-time requirement	include preregistered shorter and longer cells	identifiability check
nominal surface gap	10 µm, common to both branches	boundary-control input
finite plate size	80 µm × 80 µm	transported finite-geometry input
boundary program	static randomized cells; nominal cadence label 0.5 Hz	transfer function unmeasured
primary sequence	Ramsey-type branch preparation/recombination	design assumption
control sequences	identical-branch/sham splitter, echo, path swap, sham switch, detuned boundary	required nuisance and interaction discrimination
temperature	4 K	design target, unmeasured in integrated apparatus
pressure	1×10^-15 Pa	design target, unmeasured in integrated apparatus
registered echo residual	1×10^-4	synthetic control assumption
vibration target	5×10^-10 m s^-2 Hz^-1/2	design target
readout	1550 nm, 5 nW nominal	design assumption
polarization	circular-control pair plus linear-basis checks	electromagnetic witness
companion channel	center-of-mass energy increase	detector authority absent
settling rule	to be measured and frozen from transfer-function data	not inherited from old design

The “0.5 Hz” value is a configuration label, not permission to assume that a two-second cycle is quasistatic. No continuous modulation or ten-second settling rule is admitted until the measured boundary-to-apparatus transfer function defines a compatible sequence. Static randomized cells are the default pilot implementation.

3.1 Supersession ancestry

Design record	Geometry	Status
original symmetric-force candidate	early Casimir-force-centered architecture	`superseded_design_record`
proposal-closure architecture	75 nm radius, 20 nm separation, 0.1 s, 5 μm nominal gap, 10–4 μm commissioning ladder	`superseded_design_record`
synthetic identifiability candidate	high-mass parameter-search result	parent of current manifest
Stage-4.2C silica parent	276.3 nm radius, 160 nm separation, 0.25 s, 1.2 μm gap	`superseded_search_parent`
leading Stage-4.2M design	diamond-density, 276.3 nm radius, 250 nm separation, 0.25 s, 10 μm gap	`current_synthetic_commissioning_target`

“Frozen” means immutable within a particular run. It does not mean that an older run remains the current proposal after an explicit downstream supersession.

3.2 The dominant scale gap

The leading mass is approximately 1.86235×10^{11} Da. Matter-wave interference has recently been demonstrated for sodium clusters above 170,000 Da [6]. The proposed object is therefore about 1.0955×10^{6} times more massive than that free-particle interference benchmark. The comparison is not a universal impossibility theorem—trapped mechanical systems use different state-preparation strategies—but it prevents the manuscript from treating preparation as an ordinary checklist item.

State preparation is the first physical go/no-go:

1. verify the object's mass and internal-state distribution;
2. demonstrate coherent branch preparation at the registered separation;
3. verify the 250 ms branch hold without conditioning on successful recombination;
4. measure visibility and phase stability over the full control schedule;
5. show that the preparation procedure does not itself generate the fitted nuisance signature.

Until these steps are demonstrated, the leading 1,028-window requirement and 1,600-window planning ceiling are not actionable acquisition durations. The older 542-window value remains only a Stage-4.2C parent-design record.

4. Hypotheses separated before data

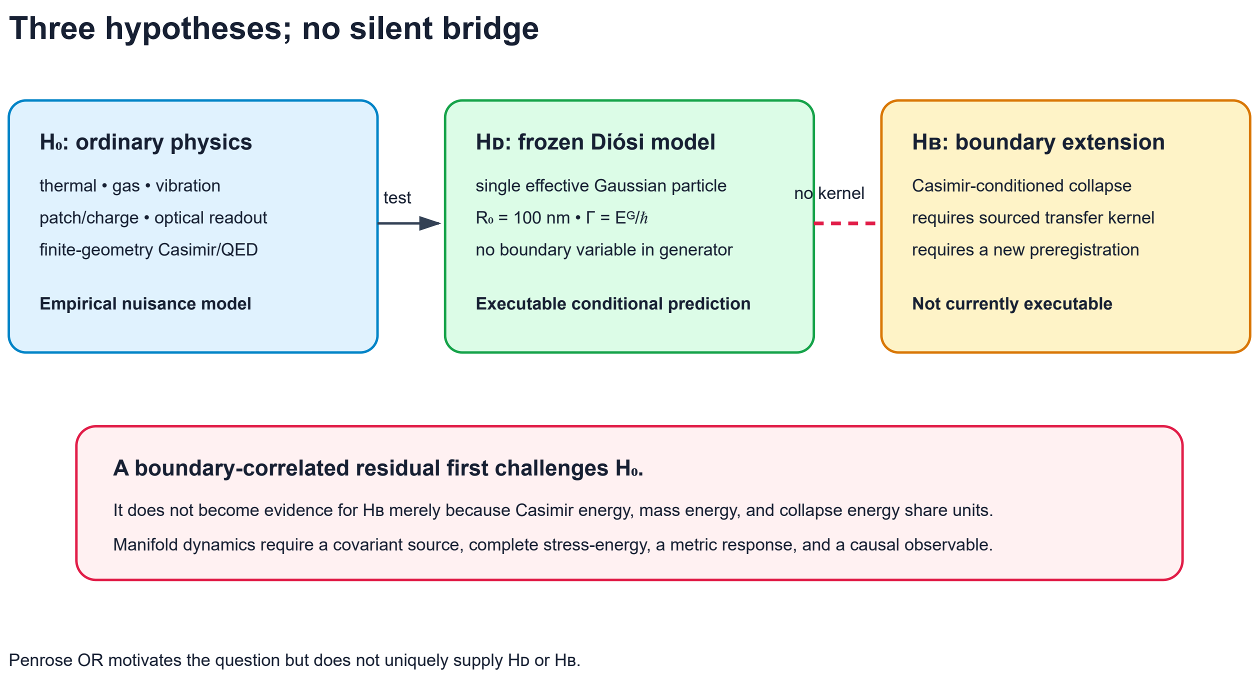


Figure 3. The ordinary-physics null, the frozen effective-particle Diósi generator, and a possible boundary-to-collapse extension are separate hypotheses. A Casimir-correlated residual cannot move from the left branch to the right branch without a registered transfer kernel and new data.

4.1 Ordinary-physics null, H0

H0 contains all registered ordinary mechanisms that can alter complex coherence:

- blackbody emission, absorption, scattering, and temperature drift;
- residual-gas collision and recoil;
- patch potentials, charge, dielectric response, and electromagnetic forces;
- vibration, tilt, support motion, and clock error;
- optical back-action, detector self-noise, and analysis leakage;
- boundary switching, material hysteresis, and finite-geometry Casimir/QED response.

The nuisance model is empirical. Textbook formulas provide priors and dimensional checks, not a substitute for measured response vectors.

4.2 Frozen Diósi hypothesis, HD

HD adds Eq. (2) to H0 with the mass, separation, R0, and hold-time law frozen before confirmatory data. The Casimir gap and modulation state do not enter the collapse generator. Under complete joint-system equivalence, standard HD therefore predicts the same collapse term in boundary-active and boundary-reference cells.

4.3 Lane C: boundary-conditioned extension, HB

HB is a future model slot in which a Casimir boundary changes collapse or coherence through an explicitly sourced transfer kernel. No such kernel is registered. The ideal Casimir energy,

$$\frac{E_{\mathrm{C}}}{A} = -\frac{\pi^2 \hbar c}{720 a^3}, \qquad P_{\mathrm{C}} = -\frac{\pi^2 \hbar c}{240 a^4}, \tag{5}$$

does not define a map from plate gap a to Γ_{D}. Real-material Lifshitz theory, finite geometry, temperature, patches, roughness, and nonequilibrium response belong first in H0.

4.4 Manifold-response hypothesis

The motivating manifold statement is a research question, not an admitted mechanism: if alternative mass configurations source incompatible gravitational geometries, a collapse-like lifetime might scale with a gravitational self-energy. The present nonrelativistic master equation tests one phenomenological realization of that scale. A spacetime claim would additionally require a covariant source, a complete apparatus stress-energy tensor, a specified dynamical metric response, and a causal observable. Those objects are absent.

4.5 Compton-frequency non-bridge

Rest energy may be written as a frequency, $\nu_{\mathrm{C}} = mc^2/h$, and the collapse energy as $\nu_G = E_G/h$. Atomic, QED, Higgs, blackbody, Zeeman, Stark, Maxwell, and gravitational benchmarks use compatible energy and frequency units. That compatibility is valuable for calibration and unit recovery but does not create a coupling between cavity modes and collapse:

$$\mathcal{K}_{\mathrm{boundary}\to\mathrm{branch/coherence}} \quad \text{is not registered.} \tag{6}$$

The same rule applies to light cones, spinors, gravitational waves, Jeans instability, Schwarzschild radii, and mass–energy equivalence. They validate parts of relativistic or quantum notation; they do not supply Eq. (6).

5. Complex-coherence estimator and identifiability

The primary datum is not a fitted scalar decay constant. Each control cell contributes a complex coherence,

$$C_k = \langle e^{i\phi} \rangle_k = \mathrm{Re}\, C_k + i \ \mathrm{Im}\, C_k. \tag{7}$$

Real and imaginary components retain phase rotations that a visibility-only analysis would discard. Let y be the stacked real vector of all components, X the matrix of registered nuisance signatures, sD the frozen Diósi signature, and Σ the block covariance. Whitening gives

$$\tilde{y} = L^{-1} y, \qquad \tilde{X} = L^{-1} X, \qquad \tilde{s}_{\mathrm{D}} = L^{-1} s_{\mathrm{D}}, \qquad L L^{\mathsf{T}} = \Sigma. \tag{8}$$

The collapse-sensitive component is the part orthogonal to the whitened nuisance span:

$$s_{\perp} = (I - P_{\tilde{X}}) \tilde{s}_{\mathrm{D}}. \tag{9}$$

5.1 Boundary--branch interaction diagnostic

The full whitened estimator remains primary, but the boundary question has a transparent four-cell projection. Let $\beta = 0, 1$ denote reference and active boundary states, and let $q = 0, 1$ denote a measured identical-branch (or preregistered sham-split) control and the separated material superposition. Normalize each cell without discarding phase,

$$\bar{C}_{\beta q}(t) = \frac{C_{\beta q}(t)}{C_{\beta q}(0)}.$$

The Stage-4.2I diagnostic freezes the complex cross-ratio

$$\mathcal{R}_\times = \frac{\bar{C}_{11}\bar{C}_{00}}{\bar{C}_{01}\bar{C}_{10}}, \qquad I_\times = -\ln|\mathcal{R}_\times|, \qquad \Phi_\times = \arg\mathcal{R}_\times. \tag{9a}$$

The registered ordinary-physics prediction is not assumed to factorize perfectly. Its measured response model supplies $\mathcal{R}_{\times,0}$, and the corrected interaction is

$$\mathcal{R}_{\times,\mathrm{corr}} = \frac{\mathcal{R}_{\times,\mathrm{obs}}}{\mathcal{R}_{\times,0}}, \quad I_{\times,\mathrm{corr}} = I_{\times,\mathrm{obs}} - I_{\times,0}, \quad \Phi_{\times,\mathrm{corr}} = \mathrm{wrap}(\Phi_{\times,\mathrm{obs}} - \Phi_{\times,0}). \tag{9b}$$

Standard H_D multiplies both separated-branch boundary cells by the same $e^{-\Gamma_\mathrm{D}t}$. That factor cancels from Eq. (9a), so this statistic is not the primary standard-Diosi test. It asks the separate question: does the boundary change branch-dependent coherence? A nonzero corrected interaction first challenges H_0 or complete-joint-system equivalence. It cannot identify a Casimir-to-collapse mechanism without the separate kernel prohibited by Eq. (6).

For log-loss cells $Y = (Y_{00}, Y_{01}, Y_{10}, Y_{11})$, the elementary contrast is $c^\mathsf{T}Y$ with $c = (1, -1, -1, 1)^\mathsf{T}$ and variance $c^\mathsf{T}\Sigma_Y c$. Stage 4.2I recovers this as the saturated four-cell special case of covariance-weighted projection; Eq. (8), with all controls and quadratures retained, remains the general estimator. If any normalized coherence is outside the registered log-coverage domain, Eq. (9a) is withheld and the raw complex cells remain authoritative.

The same campaign replaces an opaque wave-packet receipt with explicit center-of-mass custody:

$$\mathbf{d} = \mathbf{x}_B - \mathbf{x}_A, \qquad \sigma_\mathrm{CM} = \sqrt{\frac{\mathrm{tr}\,\Sigma_\mathrm{CM}}{3}}, \qquad \{R_\mathrm{sphere}, R_0, \sigma_\mathrm{CM}\} \quad \text{have distinct physical and model roles.} \tag{9c}$$

Centers, packet covariance, overlap, momentum difference, separation uncertainty, hold jitter, preparation fidelity, trajectory, and tomography provenance must agree across boundary states within frozen tolerances. The maintained Stage-4.2I fixture exercises this contract synthetically with $\sigma_\mathrm{CM} = 10\ \mathrm{nm}$; it is not a prepared-state measurement. In that boundary-independent-DP recovery, $I_{\times,\mathrm{corr}} = 1.11 \times 10^{-16}$, $\Phi_{\times,\mathrm{corr}} = -9.30 \times 10^{-19}\ \mathrm{rad}$, and the maximum interaction significance is 3.85×10^{-13}. An adversarial fixture recovers an injected (0.002) loss interaction and (0.004\ \mathrm{rad}) phase interaction, while low coherence, packet mismatch, non-positive covariance, and boundary-dependent DP fail closed. These are software-recovery results only; the four cells, packet metrology, and ordinary response are not measured.

The preregistered nominal design gates are

$$\max_j |\cos(\tilde{s}_\mathrm{D}, \tilde{x}_j)| \le 0.97, \quad \kappa(G_\mathrm{norm}) \le 100, \quad \mathrm{power} \ge 0.80, \quad \mathrm{FPR} \le 0.05. \tag{10}$$

These are design-admission gates, not proof that the modeled nuisance set is complete.

5.2 Rejected design

The initial synthetic design produced

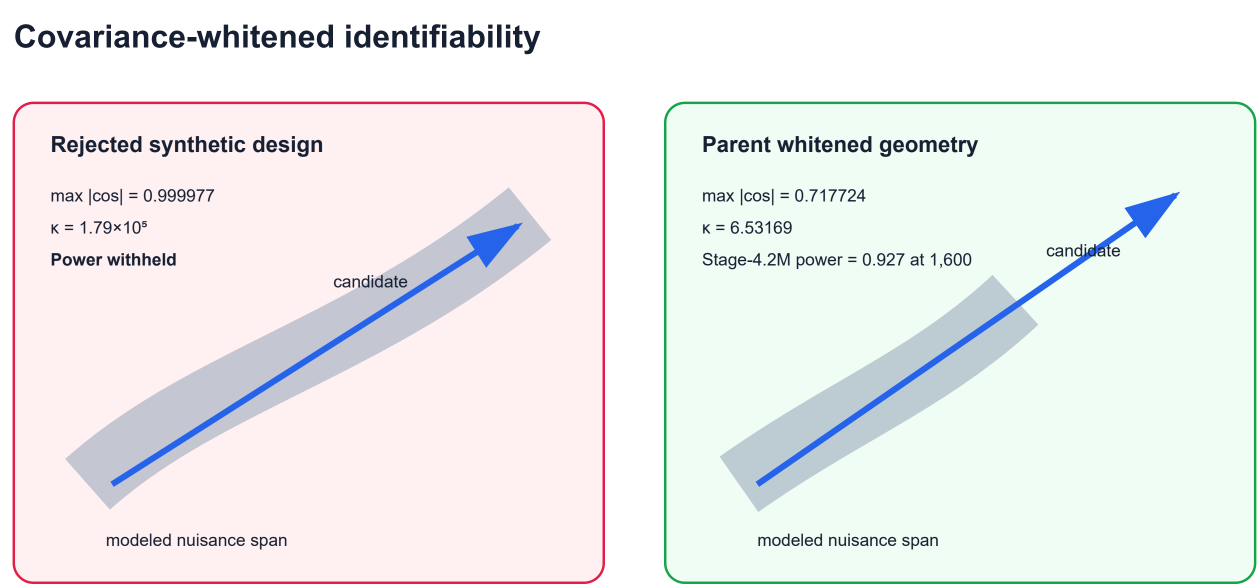
Diagnostic	Result	Gate
maximum absolute whitened cosine	0.999977	fail
normalized Gram condition number	179,104	fail
acquisition power	withheld	not meaningful

The correct result was an apparatus-design no-go. It was not a null result on the Diósi model.

5.3 Parent synthetic redesign

The Stage-4.2C parent redesign produced

Diagnostic	Nominal synthetic value	Gate
maximum absolute whitened cosine	0.717724	pass
normalized Gram condition number	6.53169	pass
forecast power	0.997858	pass in nominal world
nominal required paired windows	542	descriptive only
planned paired windows	1,600	not yet authorized



The right panel is transported, not empirical. Finite-pilot covariance, response error, missing channels, drift, tails, and selection optimism remain unstressed.

Figure 4. The first design's candidate direction lies almost inside the nuisance span. The parent redesign establishes the nominal whitened geometry later transported by Stage 4.2M. Both panels remain conditional on synthetic response vectors and an assumed covariance.

5.4 Leading Stage-4.2M design

The downstream search evaluates 200 bounded configurations under the frozen Diósi law and the same registered whitened geometry. Three pass all twelve synthetic gates. The leading point is:

Diagnostic	Leading synthetic value	Gate
maximum absolute whitened cosine	0.717724	pass; transported geometry
normalized Gram condition number	6.53169	pass; transported geometry
effective-Gaussian visibility loss at 250 ms	2.90803%	pass
conservative density-envelope loss at 250 ms	0.434903%	pass
gas-to-DP proxy ratio	0.007320	pass synthetically
echoed phase uncertainty	1.109×10^-8 rad	pass synthetically
synthetic companion SNR	8.668	pass synthetically
required paired windows	1,028	within 1,600 ceiling
forecast power at 1,600 windows	0.927386	pass

This table, not the 542-window parent result, is the current acquisition forecast. It remains a subsystem-commissioning target because its response, covariance, gas, phase, preparation, and companion inputs are transported surrogates rather than measurements.

5.5 Material-resolved ordinary complex-coherence null

Stage 4.2N replaces the transported electromagnetic scalar with an executable ordinary-response chain for the leading design. A passive specimen loss table is converted to $\epsilon(i\xi)$, a finite-geometry Green table supplies the mean branch potential, and a two-sided fluctuation--dissipation spectrum supplies phase/loss covariance. The transparent ordinary prediction is

$$C_{0,\beta}(t) = C(0)\exp[i\Phi_{\mathrm{EM},\beta}(t) - \chi_{0,\beta}(t)]\,. \tag{16a}$$

The boundary-by-superposition control is represented by the normalized four-cell ratio

$$\mathcal{R}_4 = \frac{C_{\text{active,separated}}C_{\text{reference,compact}}}{C_{\text{active,compact}}C_{\text{reference,separated}}}, \qquad \Phi_4 = \arg\mathcal{R}_4, \qquad \chi_4 = -\ln|\mathcal{R}_4|. \tag{16b}$$

This is the explicit form of the comparison already motivated by the Stage 4.2I estimator. A nonzero corrected $\mathcal{R}_4$ challenges factorization of the boundary and branch responses; it does not by itself identify collapse. The boundary-independent registered Diósi factor is identical in active and reference cells and therefore cancels from this interaction statistic. The separate Diósi exponents remain the mass--separation--time comparator for the primary coherence analysis; Stage 4.2N does not add them to the ordinary Green/FDT exponent and supplies no Casimir-to-collapse kernel.

The current synthetic fixture recovers $\Phi_4 = 0.01999999999\ \mathrm{rad}$ and $\chi_4 = 1.34165 \times 10^{-6}$, with propagated phase uncertainty $6.30 \times 10^{-5}\ \mathrm{rad}$. It also recovers zero response at zero coupling and infinite distance and a Stefan--Boltzmann relative error of 1.38×10^{-14}. These values validate executable bookkeeping only. Measured evidence and ordinary-null authority remain `not_ready`; residual attribution, collapse identification, and manifold dynamics remain `blocked`.

5.6 Why the selected power is still conditional

The response/covariance geometry used by Stage 4.2M was transported from the same synthetic world used to establish the parent candidate family. Finite-pilot covariance uncertainty was not propagated. Accordingly, 0.927386 is a conditional calculation, not a robust power claim. The 0.997858 value is retained only as the Stage-4.2C parent result. The next empirical computation must report an envelope rather than a single number.

The preregistered stress family must include:

Stress axis	Required analysis	Current status
finite-pilot covariance	nested simulation, bootstrap, or analytical propagation	not run
shrinkage/regularization	repeat whitening across declared estimators	not run
drift and non-Gaussian tails	correlated drift and heavy-tail worlds	not run
response amplitude/phase error	perturb every nuisance vector within pilot uncertainty	not run
missing nuisance	inject unregistered channels and test residual inflation	not run
branch and hold-time jitter	propagate preparation/metrology distributions	not run
shared calibration ancestry	model cross-control correlation explicitly	partially represented, not stressed
leave-one-control-out	refit after withholding each control family	not run
covariance structure	alternate preregistered block families	not run
selection optimism	repeat candidate search in independent synthetic worlds	not run

The accepted future headline has the form: “Across the registered uncertainty family, power ranged from X to Y and all design gates survived in Z% of worlds.” Until X, Y, and Z are computed, the acquisition-power claim remains provisional.

6. External constraints and scientific reach

The selected R0=100 nm point must be compared with constraints on the same model convention. Direct spontaneous-radiation searches currently provide a strong lower-bound axis for Markovian DP. The 2026 XENONnT analysis reports $R_0 > 4.9 \times 10^{-10}$ m at 90% confidence and $R_0 > 4.5 \times 10^{-10}$ m at 95% confidence in its stated Diósi convention [7]. Earlier germanium searches ruled out the natural parameter-free proposal and progressively strengthened the lower limit [8,9].

The selected $R_0 = 1.0 \times 10^{-7}$ m is about 204 times the XENONnT 90%-confidence lower bound. It is therefore not excluded by that one-sided screening comparison. This is not yet a full admission receipt: the mass-density smearing definition, normalization convention, charged-constituent radiation map, and treatment of the effective composite particle must be matched line by line before the point is called externally allowed.

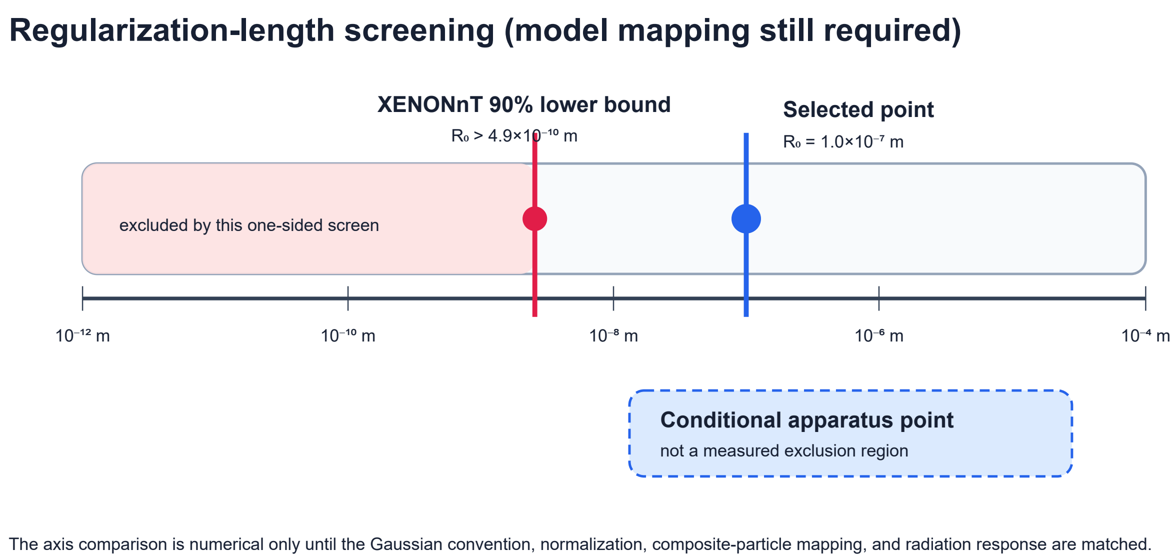


Figure 5. Logarithmic R0 screening. The XENONnT lower bound and the selected 100 nm point are shown on the same numerical axis. The shaded “apparatus” marker is conditional sensitivity, not a measured exclusion. The formal model-equivalence audit remains open.

The current reach statement is therefore limited:

- a measured null with validated sensitivity could exclude the registered effective-particle point;
- it would not exclude all regularized, dissipative, colored-noise, or Penrose-motivated models;
- a positive coherence residual could favor the frozen line shape over the registered nuisance span;
- it could not distinguish neighboring collapse generators without additional parameter and companion observables;
- a boundary-correlated excess would first challenge H0 and the joint-system equivalence assumption, not prove HB.

7. Leading-design forecast

Figure 6 shows the conditional coherence curve from Eq. (2). It contains no ordinary-decoherence band because that band must be learned from pilot data.

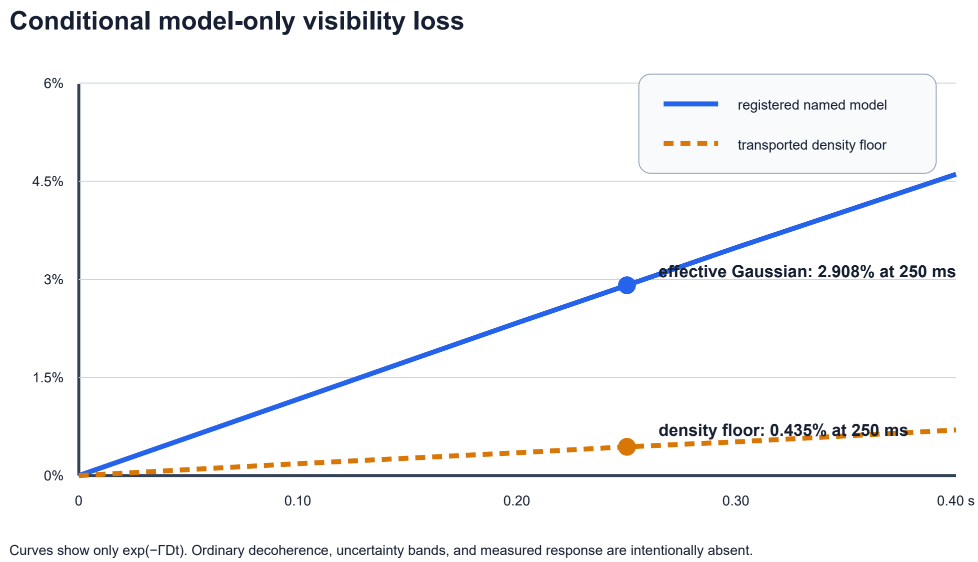


Figure 6. The registered effective-Gaussian curve reaches 2.908% loss at 250 ms, while the transported conservative density envelope reaches 0.435%. The interval is model sensitivity, not a statistical confidence band. Neither curve is a total visibility forecast.

7.1 Companion observable and interpretation ceiling

The same nondissipative generator predicts momentum diffusion

$$D_{pp} = \frac{G\hbar m^2}{12\sqrt{\pi}R_0^3} \tag{11}$$

and a center-of-mass energy increase

$$\dot{E} = \frac{3D_{pp}}{m} = \frac{G\hbar m}{4\sqrt{\pi}R_0^3}. \tag{12}$$

For the leading manifest, $\dot{E} = 3.07013 \times 10^{-40}$ W. If 100 independent samples and SNR≥5 were demanded, the algebraic maximum one-shot standard uncertainty would be

$$\sigma_{\dot{E},\text{one shot}} \leq \frac{\dot{E}\sqrt{100}}{5} = 6.14027 \times 10^{-40}\ \text{W}. \tag{13}$$

Equation (13) is not an instrument model. No registered detector demonstrates this bandwidth, calibration, independence, or noise floor. The companion observable is therefore presently a structural no-go for the paper's strongest positive interpretation.

This revision adopts the conservative third option:

> Unless an independently powered companion channel becomes feasible, a replicated DP-shaped coherence residual may be reported as unexplained model-consistent phenomenology, but not as support-eligible identification of the Diósi generator.

The project may replace the heating channel only through a preregistered derivation from the same frozen dynamics, with an instrument-level bandwidth, integration-time, noise, calibration-transfer, and cross-covariance model.

8. Empirical pilot and confirmatory decisions

8.1 Pilot first

The feasibility pilot is a hardware-characterization campaign, not a search for collapse. It must deliver provenance-bound measurements of:

1. object mass, geometry, material, coating, and internal-density model;
2. state-preparation fidelity, an explicit identical-branch or sham-split control, and separated-branch metrology;
3. center-of-mass packet centers/covariances, overlap, momentum difference, hold-time jitter, and recombination metrology in both boundary states;
4. finite-geometry electromagnetic Green response and material permittivity;
5. boundary-switching transfer functions and settling time;
6. thermal, gas, vibration, tilt, patch, charge, optical, and sensor responses;
7. raw complex-coherence covariance, including shared calibration ancestry;
8. sham, detuned, echo, and path-swap controls;
9. companion-channel feasibility or the lower interpretation ceiling;
10. blinded data custody and independent solver replication.

8.2 Pilot admission rule

$$\mathcal{G}_{\rm pilot} = \mathcal{G}_{\rm preparation} \wedge \mathcal{G}_{\rm response} \wedge \mathcal{G}_{\rm covariance} \wedge \mathcal{G}_{\rm identifiability} \wedge \mathcal{G}_{\rm power} \wedge \mathcal{G}_{\rm custody}. \tag{14}$$

The pilot is admitted to confirmatory planning only if every factor is true. The numerical identifiability gates remain Eq. (10), but power must additionally survive the uncertainty envelope in Section 5.3. If any factor fails, the result is an explicit apparatus-redesign no-go.

8.3 Confirmatory campaign

Only after pilot admission may the team freeze:

- one canonical apparatus identity;
- one mass-density representation and R0 point or preregistered grid;
- exclusions and quality-control rules;
- response and covariance ancestry;
- analysis code and randomization;
- companion interpretation ceiling;
- primary and replication sample sizes.

The confirmatory data remain blinded until completeness and custody checks pass. An independent replication is required for any positive interpretation.

8.4 Decisive outcomes

Observed outcome	Defensible inference	Not established
pilot cannot prepare the state	current apparatus infeasible	collapse model false
pilot fails cosine/conditioning gates	current design non-identifiable	collapse model false
measured null with demonstrated sensitivity	registered point disfavored/excluded at stated confidence	all DP or Penrose OR false
residual follows ordinary control	H0 response model repaired	collapse
corrected four-cell interaction is null	no resolved boundary--branch nonfactorization at demonstrated sensitivity	standard boundary-independent DP false
corrected four-cell interaction is nonzero	H0 or complete-joint-system equivalence challenged	Casimir-modified collapse without a registered kernel
boundary-correlated residual without transfer kernel	anomaly eligible for a new bridge study	Casimir-induced collapse
replicated Diósi-shaped coherence residual, no companion	unexplained model-consistent phenomenology	generator identified
replicated shape plus independent companion	tested generator gains support within registered alternatives	manifold dynamics or universal objective collapse

9. Observable-separation gate

Force, phase, coherence, heating, curvature, and collapse rate are different observables. They may be connected only by a declared model with units, parameters, provenance, and falsifiers. The governing gate is:

$$\text{observable A} \to \text{observable B} \quad \Longrightarrow \quad \{\text{transfer law, units, source, parameters, test}\}. \tag{15}$$

Equation similarity, shared use of Planck's constant, or a common energy-frequency conversion is insufficient. In particular:

- Casimir pressure is not DP self-energy;
- renormalized negative energy density is not automatically negative spacetime curvature for the complete apparatus;
- virtual-particle language is not a detector model;
- gravitational-wave interference does not imply phase coherence between Casimir and matter-wave channels;
- circular polarization is an electromagnetic control degree of freedom, not a collapse source;
- a light cone defines causal structure, not a nonrelativistic collapse generator.

10. Limitations

The main limitations are scientific, not typographic:

1. **State preparation:** the leading mass is approximately 1.10 million times the 170 kDa free-particle interference benchmark; measured packet width, overlap, trajectory, and boundary-to-boundary branch equivalence are absent.
2. **Synthetic optimization:** the selected design was evaluated in the same assumed response/covariance world in which it was found.
3. **External-bound mapping:** the selected R0 passes a numerical lower-bound screen, but the exact convention/composite mapping is not certified.
4. **Mass representation:** the 2.908% prediction is specific to one effective Gaussian particle; the transported representation envelope lowers the design floor to 0.435%, while selected-specimen homogeneous, layered, coarse-grained, and atomistic calculations are incomplete.
5. **Companion channel:** the heating signal lacks a physically credible detector receipt, lowering the maximum positive interpretation.
6. **Ordinary physics:** finite-geometry material and environmental responses are unmeasured for the integrated apparatus.
7. **Relativity:** the registered dynamics is nonrelativistic and does not compute manifold evolution.
8. **Boundary bridge:** no Casimir-to-collapse transfer kernel exists in the tested model.

The machine-readable status is equivalent to the following prose: measured evidence is not yet available; empirical pilot readiness and physical viability have not been established; collapse identification and manifold dynamics remain blocked.

11. Discussion

11.1 What is new

The useful result is not that a sufficiently massive object has a large collapse rate; that scaling is already built into the selected model. The advance is the conversion of that rate into a fail-closed experimental comparison. A candidate signal is expressed in the same complex-coherence coordinates as the measured control responses, whitened by their joint covariance, and tested for linear dependence before acquisition power is reported. The rejected design demonstrates why this order matters. A nominally nonzero loss can be statistically powerless as evidence when its direction is indistinguishable from ordinary drift or decoherence.

This approach also changes what counts as a useful negative result. Failure to prepare the state is an apparatus no-go. Failure of the cosine or conditioning gate is an identifiability no-go. A measured null becomes a model constraint only after the apparatus demonstrates sensitivity to the frozen direction. These outcomes answer different questions and should not be combined into a single “experiment failed” category.

11.2 Why retain the Casimir boundary

The Casimir boundary remains valuable even though it does not enter Eq. (2). It creates a tunable electromagnetic and material environment with strong gap, temperature, polarization, and surface dependence. That makes it an unusually demanding test of whether the ordinary-physics null is complete. Sham switching, detuning, static gap cells, path swaps, and polarization controls can reveal phase rotations or losses that might otherwise be misidentified as a mass-dependent collapse residual.

This use is scientifically asymmetric. Agreement with finite-geometry macroscopic QED improves confidence in H0; disagreement produces a boundary-response anomaly. Neither result modifies HD automatically. Only a new, independently motivated transfer law could make the boundary a parameter of a collapse generator. Keeping that bridge closed protects both sides of the experiment: ordinary boundary physics is not mislabeled as gravity, and a failure of the particular Diósi model is not mislabeled as a failure of Casimir theory.

11.3 What a positive result would mean

A coherence residual with the registered mass, separation, and hold-time dependence would be more informative than an unexplained scalar decay rate. Replication across object masses and branch geometries would make an ordinary single-channel explanation progressively less plausible. Even so, the interpretation remains conditional on the alternative set. An unmodeled gas, patch, readout, preparation, or calibration response can imitate a structured signal if it shares the same experimental schedule.

The companion channel was intended to reduce that ambiguity by demanding a second prediction from the same generator. Its forecast, however, is too small to count as an experimental discriminator without a credible instrument proposal. The present claim ceiling is therefore deliberately lower than the original protocol's aspirational ceiling. A replicated coherence residual without a companion would motivate new tests and model comparison; it would not identify objective collapse. This is a scientific consequence of the detector gap, not merely cautious wording.

11.4 What a null result would mean

At a demonstrated sensitivity, a null result can exclude the registered effective-particle point. It cannot directly exclude a rigid-sphere representation that was never converged, a different regularization length, a dissipative or colored-noise theory, or Penrose's broader lifetime motivation. The mass-density robustness campaign therefore determines how widely the null may be interpreted. If several defensible representations predict comparable losses and all are within reach, the experiment tests a family. If their predictions diverge substantially, the representation itself becomes part of the preregistered alternative set.

11.5 Priority order after this design study

The next scientific work should be ordered by the chance that it invalidates the apparatus before costly acquisition:

1. demonstrate or reject state preparation at the registered mass, separation, and hold time;
2. replace the conservative residual-gas no-go with a measured composition, pressure, and scattering/collision-veto model or redesign the apparatus;
3. complete provenance-bound layered and constituent density maps;
4. complete the exact external-bound convention and composite-particle map;
5. measure finite-geometry material and boundary transfer functions;
6. obtain pilot response vectors and covariance;
7. run the registered robustness envelope and repeat candidate selection in independent synthetic worlds;
8. either demonstrate a companion instrument or retain the lower interpretation ceiling;
9. only then freeze a blinded confirmatory campaign.

This sequence keeps the paper's methodological core useful even if the present apparatus is rejected. The estimator, hypothesis separation, packet custody, and design gates can be carried into a lower-mass or otherwise redesigned platform without preserving the current sphere as a favored physical object.

11.6 Microscopic electromagnetic closure and orientation no-go

Stage 4.2K replaces the literal "missing virtual photons" picture with the ordinary macroscopic-QED chain that the apparatus must actually calibrate. For an isotropic ground state, transition frequencies and dipole matrix elements define the polarizability, while the boundary enters through the scattering Green tensor:

$$\alpha_g(i\xi) = \frac{2}{3\hbar}\sum_n \frac{\omega_{ng}|\langle n|\hat{\mathbf{d}}|g\rangle|^2}{\omega_{ng}^2+\xi^2}, \qquad U_{\rm CP}(\mathbf{r}) = \frac{\hbar\mu_0}{2\pi}\int_0^\infty d\xi\,\xi^2\alpha_g(i\xi)\,\mathrm{Tr}\,\mathbf{G}^{(1)}(\mathbf{r},\mathbf{r};i\xi).$$

For the silica sphere, the diagnostic small-particle response is

$$\alpha_{\rm sph}(i\xi) = 4\pi\epsilon_0 R^3\frac{\epsilon(i\xi)-1}{\epsilon(i\xi)+2}.$$

Charge/current response therefore controls the Casimir lane; total mass density controls the registered Diósi lane. A common composition and geometry manifest must bind them, but the two operators are not interchangeable.

The synthetic two-oscillator recovery gives $\epsilon(0) = 3.8$ and $\alpha_{\rm sph}(0) = 1.13303\times10^{-30}$ in SI units. These are software fixtures, not measured cryogenic silica data. More importantly, this superseded-parent calculation exposed the orientation constraint used by the later redesign. At a 1.2 micrometre surface gap, a 160 nm split normal to the plane gives $\Delta U_{\rm CP} = 3.55916\times10^{-24}$ J and an accumulated 250 ms phase of 8.43746×10^{9} rad. The tangential split gives zero differential phase in the translationally invariant ideal-plane limit. Neither is the final finite-geometry forecast; their difference shows that branch orientation must be registered before residual attribution.

Phase noise also contracts averaged coherence. For Gaussian phase jitter,

$$\langle e^{i\delta\phi}\rangle = e^{-\sigma_\phi^2/2}, \qquad \chi_\phi = \frac{\sigma_\phi^2}{2}.$$

Keeping this contribution below one tenth of the registered DP exponent needs $\sigma_\phi \le 0.03464$ rad in the normal screen, corresponding to roughly 4.11×10^{-12} fractional stability relative to its phase. Echoes may cancel a static phase, but their measured rejection and residual covariance must replace this screening calculation.

Finally, the Stage-4.2J collision count cannot be promoted into a complete gas decoherence rate. That replacement requires the quantum-linear-Boltzmann kernel

$$\Gamma_{\rm gas}(\Delta\mathbf{x}) = n_g\int d^3v\,\mu(\mathbf{v})v\int d\Omega\,|f(\mathbf{q},\mathbf{v})|^2\left[1-e^{i\mathbf{q}\cdot\Delta\mathbf{x}/\hbar}\right],$$

with measured species-resolved pressure and temperature, differential scattering amplitudes, confinement geometry, momentum-transfer quadrature, and independent pressure calibration. All are currently absent. Stage 4.2K thus passes as an analytic diagnostic while finite-geometry electromagnetic closure, the QLBE environment, and residual attribution remain blocked.

11.7 Derivation ancestry: empirical-authority no-go

Stage 4.2L froze the previously missing silica-parent geometry as a design authority for its own run. In a right-handed frame whose origin is the plate center, the reference sphere is at $(0, 0, 1.476302\ \mu\mathrm{m})$, the plate normal is $\hat{\mathbf{n}} = (0, 0, 1)$, and the 160 nm branch vector is $\Delta\mathbf{x} = (160\ \mathrm{nm}, 0, 0)$. Thus $\Delta\mathbf{x} \cdot \hat{\mathbf{n}} = 0$: the registered design branch is tangential. This is now a superseded search-parent manifest, retained because its no-go result defines the requirements applied by Stage 4.2M; it was never as-built CAD, branch metrology, or plate-normal metrology.

To test the geometry pipeline before a full Maxwell solve, Stage 4.2L uses a finite rectangular-plate surrogate. The plate's solid angle and the screened retarded energy are

$$\Omega(\mathbf{r}) = \iint_A \frac{z\, dA'}{[(x'-x)^2 + (y'-y)^2 + z^2]^{3/2}}, \qquad U_{\mathrm{rect}}(\mathbf{r}) = \frac{\Omega(\mathbf{r})}{2\pi} U_{\mathrm{CP}}^{\infty}(z).$$

Analytic rectangular solid angle and 160-by-160 midpoint surface quadrature agree in branch energy to 1.15×10^{-6} relative error. The centered tangential branches have equal analytic energies, and hence zero nominal differential phase. This is a symmetry diagnostic, not a full Green tensor: finite thickness, coatings, apertures, roughness, patches, and nearby conductors remain outside the surrogate.

The useful quantity is therefore not the nominal zero but its response Jacobian. For control coordinates $\boldsymbol{\theta} = (x_c, z, \vartheta_b, \vartheta_p)$,

$$J_i = \frac{\partial \Phi_{\mathrm{EM}}}{\partial \theta_i}, \qquad \sigma_\phi^2 = \mathbf{J}\, \boldsymbol{\Sigma}_\theta \mathbf{J}^{\mathsf{T}}.$$

The frozen synthetic design tolerances return $\sigma_\phi \simeq 1.28 \times 10^4$ rad, far above the registered 0.03464 rad allowance. Considered one at a time, the surrogate requires about 5.50×10^{-14} m lateral centering or 4.39×10^{-12} rad branch/plate angular stability. These values do not assert that such stability has been measured; they reject the current reference tolerances and require a geometry, gap, modulation, echo, or mass/separation redesign before a pilot.

For residual gas, Stage 4.2L evaluates the isotropic reduction of the QLBE kernel,

$$\Gamma_{\mathrm{gas}}(d) = \sum_s n_s \bar{v}_s \sigma_s \frac{1}{2} \int_{-1}^{1} d\mu \left[1 - \mathrm{sinc}\left(\frac{m_s \bar{v}_s d \sqrt{2(1-\mu)}}{\hbar}\right)\right].$$

At 4 K and 2×10^{-11} Pa, the registered H2/He isotropic proxy gives $17.28\ \mathrm{s}^{-1}$, about 720 times the registered DP rate. Reaching one tenth of that rate would require approximately 2.78×10^{-15} Pa under the same proxy. Because measured species, differential scattering, confinement, and pressure calibration are absent, this is a redesign no-go rather than an empirical environmental forecast.

Stage 4.2L also expands the DP representation screen. For a normalized radial form factor F_a,

$$E_G^{(a)}(d, r_0) = \frac{2Gm^2}{\pi r_0} \int_0^\infty du\, e^{-u^2} |F_a(uR/r_0)|^2 \left[1 - \mathrm{sinc}(ud/r_0)\right].$$

At $r_0 = 100$ nm, the effective Gaussian, homogeneous sphere, thin shell, and a 5% shell sensitivity profile span 3.74×10^{-37} to 2.53×10^{-36} J, a factor of 6.77. This confirms material-representation dependence; it does not choose the correct representation without internal-density and coating metrology.

The current external-bound ledger is now partly closed. XENONnT reports, in the Markovian Diósi convention, $R_0 > 4.9 \times 10^{-10}$ m at 90% confidence. The frozen $R_0 = 10^{-7}$ m point is about 204 times above that scalar lower bound and is not excluded by this one-dimensional screen. A source-level likelihood and composite-representation recast remain required before calling the parameter point fully mapped.

Finally, the reference object's mass is 1.1706×10^{11} Da. The 2026 matter-wave benchmark demonstrates more than 170 kDa with 133 nm separation, so the proposed mass is about 6.89×10^{5} times larger even though the separation is only 1.20 times larger. The material, diameter, trapping, and interferometer platform also differ. State preparation therefore remains an unclosed physical-feasibility gate.

11.8 Selection of the leading bounded design

Stage 4.2M defines the paper's leading design by asking whether any point in a frozen, bounded apparatus domain can simultaneously make the registered DP signal resolvable while keeping the transported electromagnetic, gas, preparation, identifiability, power, and companion constraints within their preregistered limits. The search does not refit the DP generator, use confirmatory data, or introduce a Casimir-to-collapse transfer law.

For each candidate q, the registered decision is the conjunction

$$G_M(q) = G_{\mathrm{DP}} \wedge G_\rho \wedge G_\phi \wedge G_{\mathrm{gas}} \wedge G_{\mathrm{prep}} \wedge G_{\mathrm{bound}} \wedge G_{\mathrm{id}} \wedge G_{\mathrm{power}} \wedge G_{\mathrm{companion}}.$$

The ordinary-background quantities are transported as explicit design surrogates rather than declared measured authorities. In particular,

$$\sigma^2_{\phi,\mathrm{echo}} = \eta^2_{\mathrm{echo}} \mathbf{J}_\phi(q) \mathbf{\Sigma}_q \mathbf{J}_\phi^{\mathsf{T}}(q), \qquad \Gamma_{\mathrm{gas}}(q) = \Gamma_{\mathrm{gas,L}}\, \mathcal{T}_{P,T,R,d}(q).$$

The deterministic 200-candidate search returns three points that pass every synthetic gate. The best-ranked point uses a diamond-density sphere with radius 276.302 nm and mass 3.09251×10^{-16} kg, a 250 nm tangential branch, a 250 ms hold, a 10 micrometre gap, 4 K, and 10^{-15} Pa. Under the frozen effective-Gaussian Diósi law it gives $\Gamma_{\mathrm{DP}} = 0.118046\ \mathrm{s}^{-1}$, a 2.908% Gaussian-model loss, and a conservative density-envelope loss of 0.435%. Its transported diagnostics give $\Gamma_{\mathrm{gas}}/\Gamma_{\mathrm{DP}} = 0.00732$, echoed phase uncertainty 1.11×10^{-8} rad, maximum whitened cosine 0.7177, condition number 6.53, 1,028 required paired windows, power 0.927 at 1,600 windows, and synthetic companion SNR 8.67.

This is a bounded commissioning target, not a solved apparatus. The full finite-geometry Maxwell response, measured material spectra, measured phase covariance, species-resolved QLBE inputs, state preparation, companion detector, and exact external-bound recast remain absent. The mass is still approximately 1.10×10^{6} times the 170 kDa matter-wave benchmark. Accordingly, measured evidence remains not ready; residual attribution, collapse identification, and manifold dynamics remain blocked; and physical viability remains unevaluated.

12. Claim boundaries

This paper establishes:

- the exact model and parameter point used by the forecast;
- the failure of the first synthetic identifiability design;

- the nominal separability of a bounded redesign under stated assumptions;
- a leading synthetic design with a 2.908% effective-Gaussian prediction and a 0.435% transported density-envelope floor;
- a selected-candidate forecast of 1,028 paired windows and power 0.927 at the 1,600-window ceiling;
- a representation envelope showing that the effective-Gaussian prediction is not representation-independent;
- a transported gas screen that passes at the leading 10^{-15} Pa target while remaining unmeasured;
- an ideal-plane orientation screen showing that boundary phase and phase jitter can dominate the target residual unless geometry is frozen and empirically calibrated;
- a frozen 3D search geometry, a crosschecked finite-plate surrogate, and explicit phase-control tolerances that define the selected echo/alignment target;
- a QLBE-structured gas proxy, current scalar external-bound screen, and four-representation DP sensitivity envelope;
- a bounded 200-candidate redesign search with three synthetic-only points that pass the frozen multigate screen;
- a fail-closed pilot and confirmatory decision structure;
- the empirical and computational work still required.

This paper does not establish:

- that the apparatus can be built or the state prepared;
- that the nominal power survives model uncertainty;
- that the selected point is fully admitted by all external constraints;
- that a Casimir boundary changes a collapse rate;
- that objective collapse occurs;
- that Penrose's geometric interpretation is observed;
- that spacetime manifold dynamics have been measured or simulated.

13. Conclusion

The strongest result is methodological. A scalar loss rate that is nearly collinear with ordinary backgrounds is not an adequate collapse test. The experiment must operate in raw complex-coherence space, measure the nuisance responses and covariance, project the frozen candidate away from that span, and fail closed when the geometry is ill-conditioned.

The leading design is the three-point Stage-4.2M region, not the earlier silica reference. Its best-ranked point is a diamond-density 276.302 nm-radius sphere, 250 nm tangential split, 250 ms hold, 10 μm gap, 4 K, and 10^{-15} Pa. The frozen effective-Gaussian model predicts 2.908% loss; the transported mass-density envelope sets a 0.435% design floor. In the registered whitened space, the point retains cosine 0.7177 and condition number 6.53, requires 1,028 paired windows, and reaches forecast power 0.927 at 1,600 windows.

The earlier no-go calculations are the evidence that shaped this design. They showed that the first signature was non-identifiable, the silica parent was gas dominated at its declared pressure, normal boundary orientation created an overwhelming phase screen, centered tangential cancellation was tolerance sensitive, and the DP energy varied by a factor of 6.77 across registered mass profiles. Those results are retained as derivation ancestry rather than presented as competing current apparatuses.

The leading design is still not physically viable evidence. Its electromagnetic response, covariance, gas kernel, echo rejection, state preparation, material density, external-bound mapping, and companion detector are transported or unmeasured. It therefore authorizes measured subsystem commissioning only, not a physical pilot, confirmatory campaign, collapse claim, or manifold claim.

The integrated Stage-4.2M software path passes 62 focused tests, the 101-test Stage-4.2C-through-M campaign replay, the 179-test required GR/WARP battery, the 221-entry math registry, equation-sidecar and root-to-leaf validation, and both production builds. The fresh leading-design publication-rebase adapter run `2374` (`adapter:1d8a8433-4f7a-4f8e-81d7-19ac5f0fcbfc`) returns `PASS`, no first failure, empty deltas, certificate status `GREEN`, integrity true, and certificate SHA-256 `6e84f965957f63aad452981d2ede72e62f706d32e0a5b6b469899884e12a4e45`. That certificate establishes repository convergence only.

The appropriate next action is therefore a measured feasibility pilot. If the pilot reproduces the response geometry and survives the registered robustness family, a blinded confirmatory campaign becomes justified. If it does not, the apparatus is redesigned or rejected without converting an engineering failure into a claim about nature.

Publication declarations

Author contributions. To be completed by the human authors before submission. Repository history records the provenance of calculations and AI-assisted drafting; it does not determine scholarly authorship.

Code and data availability. The executable configurations, synthetic fixtures, reports, tests, equation-action sidecars, and receipts are maintained in the CasimirBot repository. The complete development record is preserved in the reproducibility supplement. No measured experimental dataset is reported.

Competing interests. To be declared by the human authors before submission.

Acknowledgments. To be completed before submission.

References

1. L. Diósi, "A universal master equation for the gravitational violation of quantum mechanics," *Physics Letters A* **120**, 377–381 (1987). https://doi.org/10.1016/0375-9601(87)90681-5
2. L. Diósi, "Models for universal reduction of macroscopic quantum fluctuations," *Physical Review A* **40**, 1165–1174 (1989). https://doi.org/10.1103/PhysRevA.40.1165
3. A. Bassi, K. Lochan, S. Satin, T. P. Singh, and H. Ulbricht, "Models of wave-function collapse, underlying theories, and experimental tests," *Reviews of Modern Physics* **85**, 471–527 (2013). https://doi.org/10.1103/RevModPhys.85.471
4. R. Penrose, "On gravity's role in quantum state reduction," *General Relativity and Gravitation* **28**, 581–600 (1996). https://doi.org/10.1007/BF02105068
5. R. Penrose, *The Road to Reality* (Jonathan Cape, London, 2004), Chapter 30.

6. S. Pedalino *et al.*, "Probing quantum mechanics with nanoparticle matter-wave interferometry," *Nature* **649**, 866–870 (2026). https://doi.org/10.1038/s41586-025-09917-9
7. E. Aprile *et al.* (XENON Collaboration), "Search for Spontaneous Radiation from Wave Function Collapse in the XENONnT Experiment," *Physical Review Letters* **136**, 120201 (2026). https://doi.org/10.1103/2jm3-4976
8. S. Donadi *et al.*, "Underground test of gravity-related wave function collapse," *Nature Physics* **17**, 74–78 (2021). https://doi.org/10.1038/s41567-020-1008-4
9. I. J. Arnquist *et al.* (Majorana Collaboration), "Search for Spontaneous Radiation from Wave Function Collapse," *Physical Review Letters* **129**, 080401 (2022); Erratum **130**, 239902 (2023). https://doi.org/10.1103/PhysRevLett.129.080401
10. M. Carlesso, S. Donadi, L. Ferialdi, M. Paternostro, H. Ulbricht, and A. Bassi, "Present status and future challenges of non-interferometric tests of collapse models," *Nature Physics* **18**, 243–250 (2022). https://doi.org/10.1038/s41567-021-01489-5
11. H. B. G. Casimir, "On the attraction between two perfectly conducting plates," *Proceedings of the Royal Netherlands Academy of Arts and Sciences* **51**, 793–795 (1948).
12. E. M. Lifshitz, "The theory of molecular attractive forces between solids," *Soviet Physics JETP* **2**, 73–83 (1956).
13. M. Arndt and K. Hornberger, "Testing the limits of quantum mechanical superpositions," *Nature Physics* **10**, 271–277 (2014). https://doi.org/10.1038/nphys2863